



Research paper

Cooperative event-triggered control for the multi-USVs via the formation reconstruction[☆]Guoqing Zhang^{a,b,c}, Chengqian Shi^a, Jiqiang Li^{a,*}, Xianku Zhang^a^a Navigation College, Dalian Maritime University, Dalian 116026 Liaoning, People's Republic of China^b State Key Laboratory of Maritime Technology and Safety, Dalian, 116026, Liaoning, People's Republic of China^c School of Information and Communication Engineering, Hainan University, Haikou, 570228, Hainan, People's Republic of China

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ABSTRACT

In this paper, a robust neural cooperative path following control algorithm is designed for multi-unmanned surface vehicles (USVs) to address the problems of the wreck avoidance by an utilization of the formation reconstruction mechanism and event-triggered rule. For this purpose, an artificial potential field (APF) guidance principle is developed, where can guide a local avoidance obstacle effect without affecting the global path following operation by designing a formation reconstruction mechanism. The major feature is that the problems of the local minimum and unattainable destination for the traditional APF are settled by presenting a velocity coordination strategy, ensuring a cooperative performance of the USVs while encountering the wreck obstacles. For the control module, a novel dynamic event-triggered rule is proposed by introducing a feedback function of output error, which can avoid the restriction of the fixed threshold parameters. Owing to this merit, the actuation frequency of the control law and adaptive neural parameter is reduced for saving a limited transmission resource usage. Further, the actuator failures caused by the potential factors, see for example saturation, delay and hysteresis are discussed by employing the two adaptive law, where the unknown gain-functions are free.

1. Introduction

In past decades, unmanned surface vehicle (USV) formation control has attracted considerable research interest due to their efficiency and robust adaptability, see for example environmental exploration, escort missions, and search and rescue (Zhang et al., 2022b) to name but a few. Nonetheless, when navigating in narrow waters, the presence of unexpected obstacles like wreck obstacles lead easily to unsafe behaviors (Zhang et al., 2024a). Therefore, the development of an effective formation avoidance (Cheng et al., 2021; Liu et al., 2022; Sun et al., 2023; Huang et al., 2019) is extremely meaningful.

As for the USV formation, guidance and control represent the concrete implementation of the obstacle avoidance scheme, serving as two critical subsystems within the closed-loop system. Line of sight (LOS) guidance (Wang and Ki Ahn, 2020) focuses on the tracking accuracy of the straight line, and has poor tracking effect in a curve path. The

dynamic virtual ship (DVS) guidance described in Zhang et al. (2025) can better address the above problems, and the simulation proves that the USV robustness stronger. At present, prevailing obstacle avoidance methods include behavior-based method (Zhang et al., 2021), dynamic window approach (Luo et al., 2023), velocity obstacle (Wu et al., 2020), artificial potential field (APF) (Sang et al., 2021) to name but a few. The APF has been concerned because of its intuitive concept and great obstacle avoidance effect. However, the traditional APF frequently encounters problems such as local minimum and unreachable target (Zhang et al., 2024b; Wu et al., 2023). In order to overcome these defects, a number of scholars have improved the APF. In recent years, additional potential fields (Pan et al., 2022), reinforcement learning (Li et al., 2021), genetic algorithm (Nazarahari et al., 2019), distributed guidance law (Gu et al., 2021) and other methods have been employed for improving the adaptability and robustness of the

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algorithm. These improvements provide new ideas for USVs formation obstacle avoidance.

Formation obstacle avoidance control requires the USVs to effectively pass obstacles with a reasonable formation structure (Lu et al., 2021), or to adjust the formation shape dynamically to adapt to the environment. In Sun et al. (2023), the introduction of nonlinear model predictive control and distributed control can ensure the cooperation and robustness of the formation and improve the safety of USVs passing through narrow waters by switching the triangular formation to a linear formation. Liu et al. (2024) describes an affine transformation and accurately evaluates collision risks by combining APF. However, the above control architecture often result in excessive computational load and actuator wear. Note that the leader-follower method (Huang et al., 2021b; Peng et al., 2021) is a control architecture with low computational load and simple to implement efficiently. Hence, it is necessary to take full advantage of the leader-follower approach and develop an improved leader-follower method that specifically design for formation avoidance missions.

For the control part, when USV formation performs the obstacle avoidance mission, the real-time signal in the closed loop system needs to be transmitted at all times. This process leads to the problem of excessive computational load and affects the real-time response ability of the control (Wang et al., 2023). Various control methods have been developed by scholars, e.g. sliding control (Li et al., 2018), model predictive control (Nan et al., 2024), event-triggered control (ETC) (Huang et al., 2020; Cheng et al., 2024). As a concise and efficient control method, ETC has been adopted broadly in intelligent ship systems (Zhang et al., 2017; Zhou et al., 2023; Li et al., 2024a,b). Huang et al. (2020) describes a dynamic event-triggered control (DETC) based on output, which adaptively updates the trigger threshold instead of a fixed parameter to reduce unnecessary resource usage. In Zhang et al. (2022a), a practical multiport event-triggered control is proposed, which is different from the traditional single-port ETC, to enable to reduce the computational resources from the sensor-to-controller. In the marine environment, due to the surge force and the yaw moment provided by the actuator cannot be accurately measured, actuator failure can occur frequently. The high performance control algorithms are the key techniques for the problem. In Huang et al. (2021a), the tolerant control and iterative learning control are adopted to settle the problem of the signal faults. The integration of fault-tolerant and ETC (Zhu et al., 2022; Zhang and Yang, 2020; Hu et al., 2023) is the powerful guarantee to control the closed-loop system converge to the reference signals. Therefore, it is significance to composite fault-tolerant technique and DETC.

Motivated by the above observations, a cooperative DETC algorithm is proposed for USV formation obstacle avoidance. Besides, the issues of the gain-related uncertainties, the nonlinear terms and explosion of complexity are addressed by using adaptive strategy, robust neural damping technique and dynamic surface control (DSC), which have been employed in the previous work (Zhang et al., 2022a; Li et al., 2024a). The main novelty of this paper can be summarized as twofold:

- (1) An artificial potential field (APF) guidance principle is developed for the purpose of formation reconstruction to achieve a wreck avoidance operation, such as changing the formation system into two subsystems with the same leader virtual ship. Further, a dynamic velocity coordination mechanism is presented to release the restriction of the local minimum ensuring formation coordination.
- (2) A robust neural cooperative control algorithm is designed for multi-USVs using a newly dynamic event-triggered rule and the adaptive neural technique. The control commands and the neural network weight estimators are updated only at trigger instants, which help minimize unnecessary transmission resource usage in channel of controller to actuator, especially in comparison to current methods.

2. Problem formulation and preliminaries

2.1. Mathematical model of the USV

In this part, according to the Newtonian or Lagrangian mechanics, the three degrees-of-freedom (3-DOFs) dynamic model of underactuated vehicles can be expressed as

$$\begin{cases} \dot{x}_i = u_i \cos(\psi_i) - v_i \sin(\psi_i) \\ \dot{y}_i = u_i \sin(\psi_i) + v_i \cos(\psi_i) \\ \dot{\psi}_i = r_i \\ \dot{u}_i = f_{ui}(\mathbf{v}_i) + \frac{T_{ui}(\cdot)}{m_{ui}} |n_{ai}| n_{ai} + d_{wui} \\ \dot{v}_i = f_{vi}(\mathbf{v}_i) + d_{wvi} \\ \dot{r}_i = f_{ri}(\mathbf{v}_i) + \frac{F_{ri}(\cdot)}{m_{ri}} \delta_{ai} + d_{wri} \end{cases} \quad (1)$$

$$\begin{cases} f_{ui}(\mathbf{v}_i) = \frac{m_{vi}}{m_{ui}} v_i r_i - \frac{d_{u1}}{m_{ui}} u_i - \frac{d_{u2}}{m_{ui}} |u_i| u_i - \frac{d_{u3}}{m_{ui}} u_i^3 \\ f_{vi}(\mathbf{v}_i) = -\frac{m_{ui}}{m_{vi}} u_i r_i - \frac{d_{v1}}{m_{vi}} v_i - \frac{d_{v2}}{m_{vi}} |v_i| v_i - \frac{d_{v3}}{m_{vi}} v_i^3 \\ f_{ri}(\mathbf{v}_i) = \frac{(m_{ui} - m_{vi})}{m_{ri}} u_i v_i - \frac{d_{r1}}{m_{ri}} r_i - \frac{d_{r2}}{m_{ri}} |r_i| r_i - \frac{d_{r3}}{m_{ri}} r_i^3 \end{cases} \quad (2)$$

where $\eta_i = [x_i, y_i, \psi_i]^T$ indicates the surge, sway displacement and yaw angle in the earth-fixed coordinate, respectively. $\mathbf{v}_i = [u_i, v_i, r_i]^T$ refers to the velocities of surge, sway and yaw. The nonlinear functions $f_{ui}(\mathbf{v})$, $f_{vi}(\mathbf{v})$ and $f_{ri}(\mathbf{v})$ denote the structure uncertainties in the mathematical model. d_{wui} , d_{wvi} and d_{wri} mean the oceanic disturbances induced by the sea wind and sea waves. n_i and δ_i stand for the actual inputs of surge force and yaw moment respectively. m_{ui} , m_{vi} , m_{ri} , d_{u1} , d_{u2} , d_{u3} , d_{v1} , d_{v2} , d_{v3} , d_{r1} , d_{r2} and d_{r3} mean all considered as unknown parameters, and they describe the ships' inertia, hydrodynamic damping and nonlinear damping terms. Since the gain functions $T_{ui}(\cdot)$ and $F_{ri}(\cdot)$ are employed to describe the dynamic of the propeller and rudder, and the implementation of the actuators requires positive values which must exist T_{u0} , T_{uM} , F_{u0} and F_{uM} satisfying $0 < T_{u0} < T_{ui}(\cdot) < T_{uM}$, $0 < F_{u0} < F_{ui}(\cdot) < F_{uM}$. The gain functions exhibit the form

$$\begin{aligned} T_{ui}(\cdot) &= (1 - t_{pi}) \rho_{si} D_{pi}^4 K_{Ti}(J_{pi}) \\ F_{ri}(\cdot) &= \left[\frac{-6.13 \lambda_{Ri}}{\lambda_{Ri} + 2.25} \cdot \frac{A_{Ri} (u_{Ri}^2 + v_{Ri}^2) \cos(\delta)}{L_{Ri}^2} \right] (x_{Ri} + \alpha_{Hi} x_{Hi}) \end{aligned} \quad (3)$$

where t_{pi} represents the wake fraction around the propeller, ρ_{si} is the water density, K_{Ti} denotes the thrust coefficient. α_{Hi} is the attack angle of rudder. D_{pi} denotes the propeller diameter. J_{pi} describes the progress coefficient of propeller. λ_{Ri} and A_{Ri} are the aspect ratio and rudder area respectively. L_{Ri} denotes the length between perpendiculars. u_{Ri} and v_{Ri} represent the relative surge and sway velocities of the rudder. x_{Ri} and x_{Hi} are the dimension parameters.

Furthermore, the actuator inputs n_{ai} and δ_{ai} may be faulty in practical engineering. For this purpose, the actuator faults model is presented in (4) that may exist in the main engine or the rudder servo system. It should be noted that variables $N_{uai} = |n_{ai}| n_{ai}$ and $N_{rai} = \delta_{ai}$ are actual input, variables $N_{uoi} = |n_{oi}| n_{oi}$ and $N_{roi} = \delta_{oi}$ are control order. n_{oi} and δ_{oi} are the inputs of the actuators, which are introduced as Section 4.

$$N_{iai} = k_{N_{oi}} N_{toi} + \bar{N}_{iai}, t \in [T_{N_{oi}}^b, T_{N_{oi}}^e], t = u, r \quad (4)$$

where $k_{N_{oi}}$ denotes the loss of effectiveness constant, $0 < k_{N_{oi}} \leq 1$. $\bar{N}_{iai}$ represents the bounded bias fault constant. $T_{N_{oi}}^b$ and $T_{N_{oi}}^e$ are the beginning and end time for the fault condition. The model (4) could be used to describe the following four fault modes. Detailed actuator failure models are described in Hu et al. (2023).

Control Objective: The control objective of this article is to accomplish the waypoints-based and APF obstacle avoidance mission with the ideal formation configuration, where the formation configuration can be described as follows

$$\lim_{t \rightarrow \infty} \|\eta_i(t) - \mathbf{A}_i - \eta_{ri}(t)\| \leq \varepsilon_0 \quad (5)$$

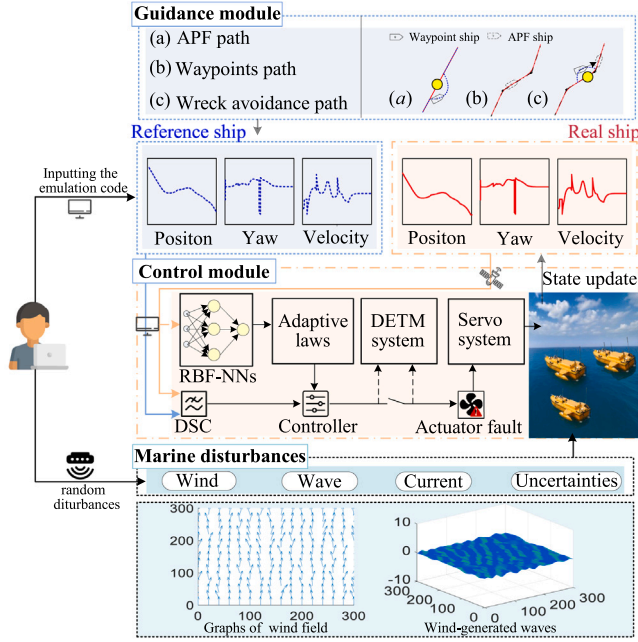


Fig. 1. The flowchart of the closed-loop system.

where $\varepsilon_0 \in \mathfrak{R}$ is a small positive constant. The coordinate transformation $\Delta_i = [\Delta x_i, \Delta y_i, \Delta \psi_i]^T \in \mathfrak{R}^3$ can be expressed as $\Delta_i = \mathbf{R}(\psi_{ri}) \mathbf{l}_i, i=1,2, \dots, N$. $\mathbf{l}_i = [\rho_{ri} \cos \lambda_{ri}, \rho_{ri} \sin \lambda_{ri}, 0]^T \in \mathfrak{R}^3$ denotes the vector determined the configuration of from the GVS-leader to GVS-followers. ρ_{ri} and λ_{ri} represent the relative distance and bearing angle between the GVS-leader and GVS-followers, respectively. $\mathbf{R}(\psi_{ri})$ given in (6) indicates the rotation of the GVS-leader. $\boldsymbol{\eta}_{ri} = [x_{ri}, y_{ri}, \psi_{ri}]^T \in \mathfrak{R}^3$ denotes the desired formation position and orientation with respect to the logic virtual ship, which will be defined in Section 3. Fig. 1 describes the flowchart of the closed-loop system under marine disturbance (Zhang et al., 2024b).

$$\mathbf{R}(\psi_{ri}) = \begin{bmatrix} \cos(\psi_{ri}) & -\sin(\psi_{ri}) & 0 \\ \sin(\psi_{ri}) & \cos(\psi_{ri}) & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad (6)$$

Assumption 1. (Zhang et al., 2022b) The reference path of the formation can be generated by a logic virtual ship, and there exists a positive constant $\xi_m \in \mathfrak{R}$ such that $\|\dot{\boldsymbol{\eta}}_{ri}(t)\| \leq \xi_m$.

Assumption 2. In the marine environment, it is assumed that the external disturbance terms satisfy: $\underline{d}_{wui} \leq d_{wui} \leq \bar{d}_{wui}, \underline{d}_{wvi} \leq d_{wvi} \leq \bar{d}_{wvi}, \underline{d}_{wri} \leq d_{wri} \leq \bar{d}_{wri}$.

2.2. RBF-NNs approximation

In the practice engineering, the radial basis function neural networks (RBF-NNs) are generally applied as an effective tool to approximate the system uncertainties (Xu and Sun, 2018). Thus, the Lemma1 will be subsequently utilized to enhance the robustness and conciseness of the control algorithm.

Lemma 1. For any given continuous nonlinear function $f(\mathbf{x})$ with $f(\mathbf{0}) = 0$, defined in a compact set $\mathbf{x} \in \Omega_{\mathbf{x}} \subset \mathbb{R}^m$, the $f(\mathbf{x})$ can be remodel by

$$f(\mathbf{x}) = \mathbf{W}^T \mathbf{S}(\mathbf{x}) + \varepsilon(\mathbf{x}), \quad \forall \mathbf{x} \in \Omega_{\mathbf{x}} \quad (7)$$

where $\mathbf{W} = [w_1, w_2, \dots, w_{\ell}]^T \in \mathbb{R}^{\ell}$ donates the NNs weight matrix. $\mathbf{S}(\mathbf{x}) = [s_1(\mathbf{x}), s_2(\mathbf{x}), \dots, s_{\ell}(\mathbf{x})]^T$ is the activation function, which can be

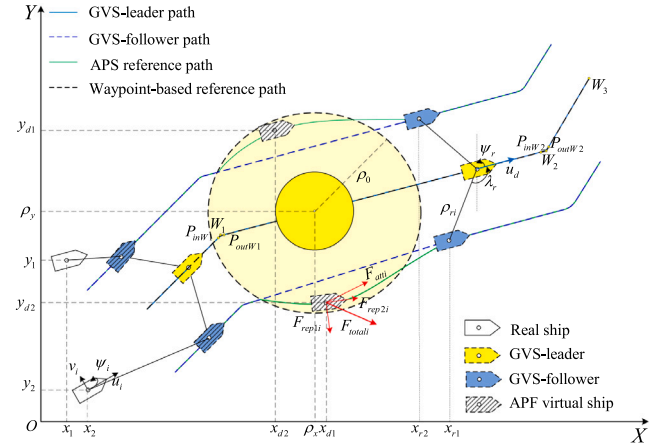


Fig. 2. Framework of formation obstacle avoidance guidance.

expressed as (8). $\varepsilon(\mathbf{x})$ is the approximation error with an unknown upper bound $\bar{\varepsilon}$, $|\varepsilon_{ii}| \leq \bar{\varepsilon}_{ii}$. $\ell > 1$ indicates the node number of NNs.

$$S_i(\mathbf{x}) = \frac{1}{\sqrt{2\pi}h_i} \exp\left(-\frac{(\mathbf{x} - \mu_i)^T(\mathbf{x} - \mu_i)}{2h_i^2}\right) \quad (8)$$

where $i = 1, \dots, \ell$. μ_i is the center of receptive field. h_i indicates the width of the activation function.

3. Obstacle collision avoidance mechanism

In this paper, the objective of the proposed obstacle avoidance mission is to provide an optimal reference path for the USVs, where the entire formation will return to the originally planned global path by changing its formation to bypass obstacles. Benefiting from the APF method, we vividly imagine the motion space of USV formation as an artificial potential field, and obstacle and guidance virtual ships (GVSS) generated by waypoints-based guidance are regarded as objects with repulsive and attractive forces on USVs, respectively.

3.1. Waypoints path navigation

In the waypoints-based guidance, the waypoints path are generated by predefined waypoints $W_1, W_2, \dots, W_n$ with $W_i = (x_i, y_i)$ as Fig. 2, which can be expressed as

$$\begin{cases} \dot{x}_r = u_r \cos(\psi_r) \\ \dot{y}_r = u_r \sin(\psi_r) \\ \dot{\psi}_r = r_r \end{cases} \quad (9)$$

Remark 1. It is difficult for the USV formation to maintain original configuration to avoid obstacles under restricted waters, which is attributed to the huge inertia of the USVs. In addition, during the obstacle avoidance process, the distances between the GVS-leader and the multiple followers are affected by APF. The excessive distance between the leader and followers can result in communication delays and decrease of control accuracy. Thus, this guidance strategy is suitable to solve the above issues caused by multiple followers.

In this paper, GVSS are divided into a GVS-leader and $n-1$ GVS-followers, $\boldsymbol{\eta}_1 = [x_{r1}, y_{r1}, \psi_{r1}]^T$ is defined as the position vector of the leader and $\mathbf{v}_1 = [u_{r1}, v_{r1}, r_{r1}]^T$ is defined as the velocities vector of the leader. Followers are generated by a leader with certain angles and distances. The formation configuration parameters ρ_{di} are formulated as

$$\rho_{di} = \sqrt{(x_{r1} - x_{ri})^2 + (y_{r1} - y_{ri})^2}. \quad (10)$$

Considering the formation configuration, USVs can obtain reasonable formation guidance signals. By adopting the path separation method, the waypoints-based planned path can be divided into straight path ($W_{i-1} \rightarrow P_{inW_i}$ and $P_{outW_i} \rightarrow W_{i+1}$) and curve path ($P_{inW_i} \rightarrow P_{outW_i}$). u_d is a positive constant determined by the ship's operator, while r_d is a time-varying value. $r_d = 0$ is established in the straight path. In the curve path, r_d can be derived as $r_d = u_d / R_{turn}$, where R_{turn} denotes the turning radius, which can be calculated by its maximum value and minimum value as

$$\phi_{i-1,i} = \arctan\left(\frac{y_i - y_{i-1}}{x_i - x_{i-1}}\right) \quad (11)$$

$$R_{turn} = \begin{cases} R_{min}, & |\Delta\phi_i| > \pi/2 \\ \frac{(R_{max} - R_{min})\Delta\phi_i}{\text{sgn}(\Delta\phi_i)\frac{\pi}{2}}, & |\Delta\phi_i| \leq \pi/2 \end{cases} \quad (12)$$

where ϕ_{i-1} is the desired heading angle between two adjacent waypoints, $\Delta\phi_i$ is the path change angle between $W_{i-1}W_i$ and W_iW_{i+1} . R_{max} and R_{min} are the maximum and minimum value.

Remark 2. The guidance part mainly serves as a global path framework for formation configuration. The reference signals generated by followers are transmitted by the leader and apply to each subsystem to facilitate the real-time tracking of the corresponding USV. The specific local obstacle avoidance process is guided by the following artificial potential field ships (APSS).

3.2. Design of the formation obstacle avoidance guidance

The APF contains two kinds of force field, namely the attractive field U_{atti} generated by GVSs and the repulsive force field U_{repi} generated by the obstacle and the total potential field. U_{totali} can be expressed as

$$U_{totali}(x, y) = U_{atti}(x, y) + U_{repi}(x, y) \quad (13)$$

Define the state of the i th APS as $\eta_{di} = [x_{di}, y_{di}, \psi_{di}]^T$ and its velocity as $v_{di} = [u_{di}, v_{di}, r_{di}]^T$. The artificial potential function can be expressed as

$$U_{atti}(\rho, v) = \frac{1}{2}k_{ap}\rho_{atti}^2 + \frac{1}{2}k_{av}v_{atti}^2 \quad (14)$$

where U_{atti} is the attractive potential fields. k_{ap} and k_{av} are the positive parameters of the attractive potential fields of positions and velocities, respectively. ρ_{atti} and v_{atti} denote the relative distance and velocity between the i th GVS and the i th APS. Taking negative gradient of U_{atti} as

$$F_{atti}(x, y) = k_{ap}\rho_{atti}\bar{n}_{api} + k_{av}v_{atti}\bar{n}_{avi} \quad (15)$$

where $\bar{n}_{api}$ and $\bar{n}_{avi}$ are the unit velocity of ρ_{atti} and v_{atti} respectively. To avoid the collision between formation and the obstacle, the repulsive potential fields are designed as

$$U_{repi}(\rho) = \begin{cases} \frac{1}{2}k_{rp}\left(\frac{1}{\rho_i} - \frac{1}{\rho_0}\right)^2\rho_{ai}^2, & \rho_i < \rho_0 \\ 0, & \text{others} \end{cases} \quad (16)$$

where U_{repi} is the repulsive potential fields generated by the island. k_{rp} is the repulsive potential fields parameter. ρ_i is the distance between the i th GVS and the obstacle. ρ_0 is the radius of the area that the repulsive potential fields effects. Take negative gradient of U_{ri} as

$$F_{repi}(x, y) = \begin{cases} -k_{rp}\left(\frac{1}{\rho_i} - \frac{1}{\rho_0}\right)\frac{\rho_{ai}^2}{\rho_i^3}\bar{n}_{ri} \\ +k_{rp}\left(\frac{1}{\rho_i} - \frac{1}{\rho_0}\right)^2\rho_{ai}\bar{n}_{api}, & \rho_i < \rho_0 \\ 0, & \text{others} \end{cases} \quad (17)$$

where F_{repi} pushes the USV formation away from the obstacle, and the closer the USV is to the obstacle, the greater the repulsive force. $\bar{n}_{ri}$ is the unit velocity of ρ_i . Therefore, the resultant force on the USV formation can be obtained as

$$F_i(x, y) = F_{atti}(x, y) + F_{repi}(x, y) \quad (18)$$

To address the issues of local minima and target inaccessibility associated with the traditional APF method, a dynamic velocity coordination model is established, which can be expressed as (19). This model is capable of resolving the formation alterations resulting from different potential field forces, while also accounting for the inability to maintain a parallel formation in the face of unknown emergencies

$$u_{ri} = \begin{cases} \min\left(\mu, \frac{\rho_i}{\chi} - k_{u_{ri}}\right)u_{ri}, & \rho_i > \rho_{v1} \\ \exp\left(\frac{\rho_i}{\chi} - k_{u_{ri}}\right)u_{ri}, & \rho_i < \rho_{v2} \\ u_{ri}, & \text{others} \end{cases} \quad (19)$$

where u_{ri} is the guidance signal of desired speed. μ , χ , $k_{u_{ri}}$, ρ_{v1} , ρ_{v2} are the positive parameters. Under the influence of total force, the desired paths by the APSs are obtained as

$$\begin{cases} \dot{x}_{di} = u_{di} \cos \psi_{di} \\ \dot{y}_{di} = u_{di} \sin \psi_{di} \\ \dot{\psi}_{di} = \theta_{Fi} \end{cases} \quad (20)$$

where θ_{Fi} is the USVs' yaw angle in the next moment as shown in Fig. 2. Therefore, the desired attitude form USVs to APSs can be derived as

$$\psi_{ri} = \frac{\pi}{2} [1 - \text{sgn}(x_{di} - x_i)] \text{sgn}(y_{di} - y_i) + \arctan\left(\frac{y_{di} - y_i}{x_{di} - x_i}\right) \quad (21)$$

where $\psi_{ri} \in (-\pi, \pi)$ is the desired attitude form USVs to APSs. The tracking error of heading angle can be obtained by $\psi_{ei} = \psi_{ri} - \psi_i$.

Remark 3. According to the distance between APSs and GVSs, the APS speed is dynamically adjusted to keep the same distance at all times. In the process of obstacle avoidance task, maintaining the sub-formation of the two groups sharing the GVS-leader can simplify the guidance structure and improve the communication efficiency, collaborates to complete the obstacle avoidance maneuvering.

Remark 4. The proposed scheme has important application prospects for safe navigation in USV formation. By planning for emergency scenarios see for example wrecks and oil spills accident, the navigation hazards are reduced. In addition, the introduction of a speed coordination mechanism enables rapid formation reconfiguration after the USV formation responds to an emergency.

4. The adaptive event-triggered formation control

4.1. Control design

Step 1: At this step, the kinematic control law will be designed for USVs. For this end, The tracking errors of position and heading angle can be described as

$$\begin{aligned} x_{ei} &= x_{di} - x_i, y_{ei} = y_{di} - y_i \\ z_{ei} &= \sqrt{x_{ei}^2 + y_{ei}^2}, \psi_{ei} = \psi_{ri} - \psi_i. \end{aligned} \quad (22)$$

As described in (1) and (22), the derivative of the error z_{ei} and ψ_{ei} have been proved in past work (Zhang et al., 2024b).

To stabilize the position the yaw error z_{ei} , ψ_{ei} , one can design virtual control law α_{ui}, α_{ri} .

$$\begin{aligned} \alpha_{ui} &= \frac{\dot{x}_{di} \cos(\psi_{di}) + \dot{y}_{di} (\sin \psi_{di}) + k_{z_{ei}}(z_{ei} - \delta_{\Delta i}) - v_i \sin(\psi_{ei})}{\cos(\psi_{ei})} \\ \alpha_{ri} &= k_{\psi_{ei}} \psi_{ei} + \dot{\psi}_{ri} \end{aligned} \quad (23)$$

where the $k_{z_{ei}}$ and $k_{\psi_{ei}}$ are both positive constants. In order to avoid the exponential explosion of second-order derivation, DSC technology (Zhang et al., 2022a) is introduced and first-order nonlinear filters are applied to

$$\epsilon_{ii} \dot{\beta}_{ii}(t) + \beta_{ii}(t) = \alpha_{ii}(t) \quad (24)$$

where ϵ_{ii} is the time constant, $i = u, r$. Filtered variable $\beta_{ii}(0) = \alpha_{ii}(0)$, kinetics errors $\iota_{ei} = \beta_{ii} - \iota_i$ and derivative of DSC error y_{ii} is $y_{ii} = -\frac{y_{ii}}{\epsilon_{ii}} + O_{ii}(\cdot)$, $|O_{ii}(\cdot)| \leq M_{ii}$, where the M_{ii} are both positive constant.

Step 2: Combining with Lemma 1, the derivatives of kinetics errors u_{ei}, r_{ei} can be obtained as

$$\begin{aligned}\dot{u}_{ei} &= \dot{\beta}_{ui} - \mathbf{W}_{ui}^T \mathbf{S}_{ui}(\mathbf{v}) - \frac{T_{ui}(\cdot)}{m_{ui}} N_{ui} - d_{wui} - \epsilon_{ui} \\ \dot{r}_{ei} &= \dot{\beta}_{ri} - \mathbf{W}_{ri}^T \mathbf{S}_{ri}(\mathbf{v}) - \frac{F_{ri}(\cdot)}{m_{ri}} N_{ri} - d_{wri} - \epsilon_{ri}\end{aligned}\quad (25)$$

Based on Lemma 1, the NNs are employed to approximate the model uncertainty $f_i(x)$. To decrease the unnecessary computation load and the wear of the actuators, the dynamic event-triggered mechanism is indicated with a time-varying threshold. Accordingly, the variable $N_{oi}(t)$ could be designed as

$$\begin{aligned}N_{oi}(t) &= N_{wii}(t_{ki}^t) \quad \forall t \in [t_{ki}^t, t_{(k+1)i}^t) \\ t_{(k+1)i}^t &= \inf \{t > t_{ki}^t \mid |e_i(t)| \geq \varpi_{oi} |N_{oi}| + d_{ii}\}\end{aligned}\quad (26)$$

where N_{wii} is an intermediate continuous control input. $e_{ii}(t) = N_{wii}(t) - N_{oi}(t)$ is the error variable. ϖ_{oi} and d_{ii} are the design parameters satisfying $\varpi_{oi} = \max(\tanh|z_{ei}|, 0)$, $\varpi_{ori} = \max(\tanh|\psi_{ei}|, 0)$, $d_{ii} = 0.5d_{ii}$ and $d_{ii} > 0$. When the mechanism is triggered, the event-triggered input $N_{oi}(t_{(k+1)i}^t) = N_{wii}(t_{(k+1)i}^t)$ would be transmitted immediately to the actuators as the control order. Apparently, $e_{ii}(t)$ could be derived as

$$N_{wii}(t) - N_{oi}(t) < \vartheta_{1i}^t(t) \varpi_{oi}(t) N_{oi} + \vartheta_{2i}^t(t) d_{ii} \quad (27)$$

It is noted that two cases can be discussed as

$$\begin{cases} \vartheta_{1i}^t(t) = \vartheta_{2i}^t(t) = \vartheta_{ii}(t) \quad \vartheta_{ii}(t) \geq 0 \\ \vartheta_{1i}^t(t) = \vartheta_{ii}(t), \vartheta_{2i}^t(t) = -\vartheta_{ii}(t) \quad \vartheta_{ii}(t) < 0 \end{cases} \quad (28)$$

where $\vartheta_{ii}(t)$ is any number satisfying $\vartheta_{ii}(t) \in [-1, 1]$. Furthermore, $N_{oi}(t)$ can be expressed as

$$N_{oi}(t) = \frac{N_{wii}(t)}{1 + \vartheta_{1i}^t(t) \varpi_{oi}} - \frac{\vartheta_{2i}^t(t) d_{ii}}{1 + \vartheta_{1i}^t(t) \varpi_{oi}} \quad (29)$$

Considering the actuator fault model and the event-triggered design, the kinetics equations can be further expressed as

$$\begin{aligned}i_{ei} &= \dot{\beta}_{ii} - \mathbf{W}_{ii}^T \mathbf{S}_{ii}(\mathbf{v}) - \frac{\Pi_{ii}(\cdot) k_{N_{oi}}}{m_{ii}(1 + \vartheta_{1i}^t(t) \varpi_{oi})} N_{wii} \\ &+ \frac{\Pi_{ii}(\cdot) k_{N_{oi}}}{m_{ii}(1 + \vartheta_{1i}^t(t) \varpi_{oi})} \vartheta_{2i}^t(t) d_{ii} - \frac{\Pi_{ii}(\cdot)}{m_{ii}} \bar{N}_{iai} - d_{wii} - \epsilon_{ii}\end{aligned}\quad (30)$$

In (30), the term $(\Pi_{ii}(\cdot) k_{N_{oi}}) / (m_{ii}(1 + \vartheta_{1i}^t(t) \varpi_{oi}))$ is the unknown and positive gain functions, $\Pi = T, F$. In order to compensate the gain uncertainty, the corresponding variables λ_{ii} is constructed to facilitate the following design respectively, that is, $\lambda_{ii} = \left[(\Pi_{ii}(\cdot) k_{N_{oi}}) / (m_{ii}(1 + \vartheta_{1i}^t(t) \varpi_{oi})) \right]^{-1}$. The intermediate control input N_{wii} , the control input n_{oi} and δ_{oi} can be expressed as, and the immediate control law $\alpha_{\zeta i}, \zeta = n, \delta$ is designed as

$$\begin{aligned}N_{wii} &= \hat{\lambda}_{ii} \alpha_{\zeta i} \\ n_{oi} &= \text{sgn}(N_{uoi}) \sqrt{|N_{uoi}|}, \delta_{oi} = N_{roi}\end{aligned}\quad (31)$$

$$\alpha_{\zeta i} = k_{\iota_{ei}} \iota_{ei} + m_{ii} \dot{\beta}_{ii} - \hat{\mathbf{W}}_{ii}^T \mathbf{S}_{ii}(\mathbf{v}) \quad (32)$$

where $k_{\iota_{ei}}$ are the control parameters. The error of the gain related adaptive law λ_{ii} is expressed as $\tilde{\lambda}_{ii} = \hat{\lambda}_{ii} - \lambda_{ii}$. The adaptive laws and the

Algorithm 1: DETC algorithm process.

Input: Variables $N_{ui}(t)$ and $\hat{\mathbf{W}}_{ii}(t)$
Output: Variables $N_{ui}(t_{ki}^t)$ and $\hat{\mathbf{W}}_{ii}(t_{ki}^t)$

```

1 begin
2    $N_{oi}(t) = N_{wii}(t_{ki}^t)$ ,  $\hat{\mathbf{W}}_{ii} = \hat{\mathbf{W}}_{ii}(t_{ki}^t)$ 
3   • Calculate control law (32).
4   • Calculate adaptive law (33).
5   for  $t = 0 : 0.01 : \text{Timeall}$  do
6     if  $t > 0$  then
7       if  $|e_i(t)| \geq \varpi_{oi} |N_{oi}| + d_{ii}$  then
8         flagEvent=0
9       else
10        flagEvent=1
11      end if
12    else
13      flagEvent=1
14    end if
15    if flagEvent = 1 then
16       $N_{oi}(t) = N_{wii}(t)$ ,  $\hat{\mathbf{W}}_{ii} = \hat{\mathbf{W}}_{ii}(t)$ 
17    else
18       $N_{oi}(t) = N_{wii}(t_{ki}^t)$ ,  $\hat{\mathbf{W}}_{ii} = \hat{\mathbf{W}}_{ii}(t_{ki}^t)$ 
19    end if
20  end for
21 end
```

update law of NNs estimator are derived as

$$\begin{aligned}\dot{\hat{\lambda}}_{ii} &= \Gamma_{\hat{\lambda}_{ii}} \left[\alpha_{\zeta i} \iota_{ei} - \sigma_{\hat{\lambda}_{ii}} (\hat{\lambda}_{ii} - \hat{\lambda}_{ii}(0)) \right] \\ \dot{\hat{\mathbf{W}}}_{ii} &= \Gamma_{\hat{\mathbf{W}}_{ii}} \left[\mathbf{S}_{ii}(\mathbf{v}) \iota_{ei} - \sigma_{\hat{\mathbf{W}}_{ii}} (\hat{\mathbf{W}}_{ii} - \hat{\mathbf{W}}_{ii}(0)) \right]\end{aligned}\quad (33)$$

where $\Gamma_{\hat{\lambda}_{ii}}$, $\Gamma_{\hat{\mathbf{W}}_{ii}}$, $\sigma_{\hat{\lambda}_{ii}}$ and $\sigma_{\hat{\mathbf{W}}_{ii}}$ are the positive constants. $\hat{\lambda}_{ii}(0)$ and $\hat{\mathbf{W}}_{ii}(0)$ denote the initial value of $\hat{\lambda}_{ii}$ and $\hat{\mathbf{W}}_{ii}$. It is noted that $\hat{\mathbf{W}}_{ii}$ at the triggering point $t = t_{ki}^t$ can be jumped as $\hat{\mathbf{W}}_{ii}(t_{(k+1)i}^t) = \hat{\mathbf{W}}_{ii}(t_{ki}^t) + \dot{\hat{\mathbf{W}}}_{ii}(t_{ki}^t) * (t_{(k+1)i}^t - t_{ki}^t)$, where t_{ki}^t is the updating time of the NNs weight estimator. Whenever the event-triggered condition is satisfied, the time instant will be marked as $t_{(k+1)i}^t$ and the corresponding $\hat{\mathbf{W}}_{ii}(t_{(k+1)i}^t)$ will be transmitted to the network for real time updates and decisions. For all the time $t \in [t_{ki}^t, t_{(k+1)i}^t)$, the NNs weight estimator will remain unchanged as a constant, i.e., $\hat{\mathbf{W}}_{ii} = \hat{\mathbf{W}}_{ii}(t_{ki}^t)$.

Remark 5. According to (30), The control inputs n_{oi} and δ_{oi} generated by DETC are usually the stepped-type, which are transmitted from the controllers to the actuators. The servo system can constrain the main engine and rudder device through amplitude saturation and rate saturation and adopt the first-order filter to smooth mutation to convert this discrete signal into a continuous signal, the actual input n_{ai} and δ_{ai} . When actuators fail, the control inputs n_{oi} and δ_{oi} will be loss. Therefore, the incorporation of fault-tolerant techniques enhances the robustness of the system.

4.2. Stability analysis

In this subsection, the stability of the proposed control system has been analyzed in Theorem 1.

Theorem 1. Consider the USV formation closed-loop control system contains USVs dynamic system (1) (2) under Assumptions 1–2, with event-triggered rules (26), controllers (31)(32), adaptive parameters and NNs weight estimators (33). For the initial conditions satisfying $(z_{ei} - \delta_{\Delta i})^2(0) + \psi_{ei}^2(0) + y_{ii}^2(0) + \iota_{ei}^2(0) + \tilde{\lambda}_{ii}^2(0) + \tilde{\mathbf{W}}_{ii}^2(0) \leq 2\Delta$, where Δ is any constant

such that $\Delta > 0$, all signals in closed-loop control system can be guaranteed SGUUB by adjusting the control parameters appropriately.

Proof. Choose the Lyapunov function as

$$V = \frac{1}{2} \sum_{i=1}^N (z_{ei} - \delta_{\Delta i})^2 + \frac{1}{2} \sum_{i=1}^N \psi_{ei}^2 + \frac{1}{2} \sum_{i=1}^N \sum_{l=u,r} (l_{ei}^2 + y_{ii}^2 + \sum_{i=1}^N \Gamma_{\mathbf{W}_{ii}}^{-1} \tilde{\mathbf{W}}_{ii}^T \tilde{\mathbf{W}}_{ii} + \sum_{i=1}^N \frac{\Pi_{ii}(\cdot) k_{N_{oi}}}{m_{ii}(1 + \vartheta_{li}^t(t) \varpi_{oi})} \Gamma_{\lambda_{ii}}^{-1}(\cdot) \tilde{\lambda}_{ii}^2) \quad (34)$$

Further,

$$\dot{V} = \sum_{i=1}^N (z_{ei} - \delta_{\Delta i}) \dot{z}_{ei} + \sum_{i=1}^N \psi_{ei} \dot{\psi}_{ei} + \sum_{i=1}^N \sum_{l=u,r} (l_{ei} \dot{l}_{ei} + y_{ii} \dot{y}_{ii} + \sum_{i=1}^N \Gamma_{\mathbf{W}_{ii}}^{-1} \tilde{\mathbf{W}}_{ii}^T \dot{\tilde{\mathbf{W}}}_{ii} + \sum_{i=1}^N \Gamma_{\lambda_{ii}}^{-1} \frac{\Pi_{ii}(\cdot) k_{N_{oi}}}{m_{ii}(1 + \vartheta_{li}^t(t) \varpi_{oi})} \tilde{\lambda}_{ii} \dot{\tilde{\lambda}}_{ii}) \quad (35)$$

Combining with the Young's inequality, the following inequalities can be obtained as

$$(z_{ei} - \delta_{\Delta i}) \dot{z}_{ei} \leq -\left(k_{z_{ei}} - \frac{1}{2}\right) (z_{ei} - \delta_{\Delta i})^2 + u_{ei}^2 + y_{ui}^2 \quad (36)$$

$$\psi_{ei} \dot{\psi}_{ei} \leq -\left(k_{\psi_{ei}} - \frac{1}{2}\right) \psi_{ei}^2 + r_{ei}^2 + y_{ri}^2$$

By submitting (32)–(33) into (25), yielding

$$l_{ei} \dot{l}_{ei} \leq -k_{l_{ei}} l_{ei}^2 - l_{ei} \tilde{\mathbf{W}}_{ii}^T \mathbf{S}_{ii}(\mathbf{v}) - \frac{1}{\lambda_{ii}} \tilde{\lambda}_{ii} l_{ei} \alpha_{\zeta i} + \frac{1}{\lambda_{ii}} \vartheta_{2i}^t(t) d_{ii} l_{ei} + l_{ei} \left(\frac{\Pi_{ii}(\cdot) \bar{N}_{rai}}{m_{ii}} + \bar{d}_{wri} + \bar{\varepsilon}_{ii} \right) \quad (37)$$

As for the effect of actuator faults, RBF-NNs approximation error and the disturbance are denoted by using variables Φ_{ii} .

$$\Phi_{ii} = \frac{\Pi_{ii}(\cdot) k_{N_{oi}} \vartheta_{2i}^t(t) d_{ii} l_{ei}}{m_{ii}(1 + \vartheta_{li}^t(t) \varpi_{oi})} + l_{ei} \left(\frac{\Pi_{ii}(\cdot) \bar{N}_{rai}}{m_{ii}} + \bar{d}_{wri} + \bar{\varepsilon}_{ii} \right) \leq \frac{1}{2} l_{ei}^2 + \Phi_{ii}^2 \quad (38)$$

where $\Phi_{ii} = \left[\frac{\Pi_{ii}(\cdot) k_{N_{oi}} \vartheta_{2i}^t(t) d_{ii}}{m_{ii}(1 + \vartheta_{li}^t(t) \varpi_{oi})} + \left(\frac{\Pi_{ii}(\cdot) \bar{N}_{rai}}{m_{ii}} + \bar{d}_{wri} + \bar{\varepsilon}_{ii} \right) \right]$, Similarly, submitting (33) into $\tilde{\lambda}_{ii} \dot{\tilde{\lambda}}_{ii}$ and $\tilde{\mathbf{W}}_{ii}^T \dot{\tilde{\mathbf{W}}}_{ii}$. The computational procedure has been shown in detail in Zhang et al. (2024b).

Submitting (35)–(38) to (34), we have

$$\dot{V} \leq -\kappa V + \varphi \quad (39)$$

where $\kappa = 2 \sum_{i=1}^N \sum_{l=u,r} \sum_{\zeta=n,\delta} \min \left\{ \left(k_{z_{ei}} - 1 \right), \left(k_{\psi_{ei}} - \frac{1}{2} \right), \left(k_{l_{ei}} - \frac{1}{2} \right), \left(1/\epsilon_{ii} - M_{ii}^2/4a_i \right), \sigma_{\lambda_{ii}}/2\lambda_{ii}, \sigma_{\mathbf{W}_{ii}}/2 \right\}$. And

$\varphi = \sigma_{\lambda_{ii}} (\lambda_{ii} - \hat{\lambda}_{ii}(0))^2 / 2\lambda_{ii} + 4a_i + \sigma_{\mathbf{W}_{ii}} (\mathbf{W}_{ii} - \mathbf{W}_{ii}(0))^2 / 2 + \Phi_{ii}^2$, where a_i is positive constant. The proposed event-triggered mechanism effectively reduces the computational burden and actuator wear between the controller and the actuator, and we should also consider avoiding Zeno behavior. Zeno behavior means that the system has to update the system state indefinitely in a finite amount of time, which is obviously impossible in practice. This phenomenon causes the control law to be calculated indefinitely, which is generally impossible to achieve with physical actuators. Therefore, We prefer that the system update interevent interval be greater than a strictly positive parameter T_{min} , that is, minimum interevent interval. In the following, we will illustrate the effectiveness of the algorithm through minimum interevent interval.

Considering (31) and (32), it is easy to note that $\dot{N}_{ai}$ is a function of all the bounded state signals in the closedloop system. So there must exist a constant $T_{mini} > 0$ such that $|\dot{N}_{ai}| < T_{mini}$. Furthermore, in

the interevent interval, one can derive (40) from the event-triggered mechanism (26). $e_{ii}(t) = N_{ai}(t) - N_{oi}(t)$ is the error variable. Assuming $e_{ii}(t_{ki}^t) = 0$, the maximum value of $e_{ii}(t)$ can be obtained as (41). Therefore, there must exist $t_{(k+1)i}^t - t_{ki}^t \geq (\varpi_{oi} |N_{oi}| + d_i) / T_{mini}$. That is, the minimum interevent interval $t_{ki}^m \geq (\varpi_{oi} |N_{oi}| + d_i) / T_{mini}$. The Zeno behavior could be effectively excluded in the proposed event-triggered mechanism.

$$\frac{d}{dt} |e_{ii}| = \text{sign}(e_{ii}) \dot{e}_{ii} \leq |\dot{N}_{ai}| \quad (40)$$

$$\lim_{t \rightarrow t_{(k+1)i}^t} e_{ii}(t) = \varpi_{oi} |N_{oi}| + d_i \quad (41)$$

Remark 6. SUGUB is concerned with the ultimate boundedness of the system within a finite region of the system state space. In addition, when USV performs the path following task, there exit non-zero disturbances due to the effect of sea wind and the wave, so it is practical to prove SGUUB.

5. Numerical experiment

In this section, two numerical examples are provided, which contain path following of wreck avoidance example and comparison example.

5.1. Path following of wreck avoidance mission

The USV formation in this paper is considered to be composed of four follower ships and a virtual leader ship. The proposed formation obstacle avoidance guidance and the control strategy are adopted to carry out the formation obstacle avoidance operation, and main results are shown in Fig. 3. The central position of the obstacle is (−200 m, 1500 m), and its radius of influence range is 720 m. The waypoints-based planned path is generated by the waypoints $W_1(550 \text{ m}, 0 \text{ m})$, $W_2(550 \text{ m}, 700 \text{ m})$, $W_3(-1500 \text{ m}, 2300 \text{ m})$, $W_4(1200 \text{ m}, 1000 \text{ m})$. The proper control parameters are set as:

$$\begin{aligned} k_{ze1} &= 0.273, k_{ze2} = 0.27, k_{ze3} = 0.27, k_{ze4} = 0.272 \\ k_{\psi e1} &= 0.35, k_{\psi e2} = 0.30, k_{\psi e3} = 0.41, k_{\psi e4} = 0.42 \\ k_{ue1} &= 2.01, k_{ue2} = 2.10, k_{ue3} = 2.11, k_{ue4} = 2.11 \\ k_{re1} &= 0.26, k_{re2} = 0.26, k_{re3} = 0.26, k_{re4} = 0.26 \\ \Gamma_{\lambda_{ui}} &= 12, \Gamma_{\lambda_{ri}} = 0.5, \sigma_{\lambda_{ui}} = 1.8, \sigma_{\lambda_{ri}} = 0.3 \\ d_{ui} &= d_{ri} = 0.1, \epsilon_{ui} = \epsilon_{ri} = 0.01 \end{aligned} \quad (42)$$

The initial states of the USVs are set as: $\boldsymbol{\eta}_1(0) = [x_1(0), y_1(0), \psi_1(0), u_1(0), v_1(0), r_1(0), x_{1d}(0), y_{1d}(0), \psi_{1d}(0)]^T = [540 \text{ m}, -20 \text{ m}, 45 \text{ deg}, 7 \text{ m/s}, 0 \text{ m/s}, 0 \text{ deg/s}, 550 \text{ m}, 0 \text{ m}, 180 \text{ deg}]^T$, $\boldsymbol{\eta}_2(0) = [x_2(0), y_2(0), \psi_2(0), u_2(0), v_2(0), r_2(0), x_{2d}(0), y_{2d}(0), \psi_{2d}(0)]^T = [300 \text{ m}, -40 \text{ m}, 45 \text{ deg}, 0 \text{ m/s}, 0 \text{ m/s}, 0 \text{ deg/s}, 350 \text{ m}, 0 \text{ m}, 90 \text{ deg}]^T$, $\boldsymbol{\eta}_3(0) = [x_3(0), y_3(0), \psi_3(0), u_3(0), v_3(0), r_3(0), x_{3d}(0), y_{3d}(0), \psi_{3d}(0)]^T = [800 \text{ m}, -30 \text{ m}, 135 \text{ deg}, 0 \text{ m/s}, 0 \text{ m/s}, 0 \text{ deg/s}, 750 \text{ m}, 0 \text{ m}, 90 \text{ deg}]^T$, $\boldsymbol{\eta}_4(0) = [x_4(0), y_4(0), \psi_4(0), u_4(0), v_4(0), r_4(0), x_{4d}(0), y_{4d}(0), \psi_{4d}(0)]^T = [980 \text{ m}, -25 \text{ m}, 135 \text{ deg}, 0 \text{ m/s}, 0 \text{ m/s}, 0 \text{ deg/s}, 950 \text{ m}, 0 \text{ m}, 90 \text{ deg}]^T$. The surge speed of GVS-leader ship is chosen as $u_d = 3.5 \text{ m/s}$, and the configuration vectors for each vehicles are given as $\mathbf{I}_1 = [200 \cos(\pi/2), 200 \sin(\pi/2), 0]^T$, $\mathbf{I}_2 = [0, 1, 0]^T$, $\mathbf{I}_3 = [200 \cos(3\pi/2), 200 \sin(3\pi/2), 0]^T$, $\mathbf{I}_4 = [400 \cos(\pi/2), 400 \sin(\pi/2), 0]^T$. The main engine speed n and rudder angle δ in the algorithm should be set to a reasonable amplitude limit, i.e., $n \in [0 \text{ rpm}, 100 \text{ rpm}]$ and $\delta \in [-35 \text{ deg}, 35 \text{ deg}]$.

Figs. 3–7 present the main results of the proposed algorithm under the marine environment. Fig. 3 describes the planar trajectory of the USV formation in the wreck avoidance path following mission. Compared with the existing literatures, when wreck accident is detected, the APSs leave the trajectory of GVSs at an angle that enables safe obstacle avoidance. In path following mode, the APSs can follow the trajectories of the GVSs. The real vessels follow the trajectories of the APSs in any mode. The tracking error of each USV and its

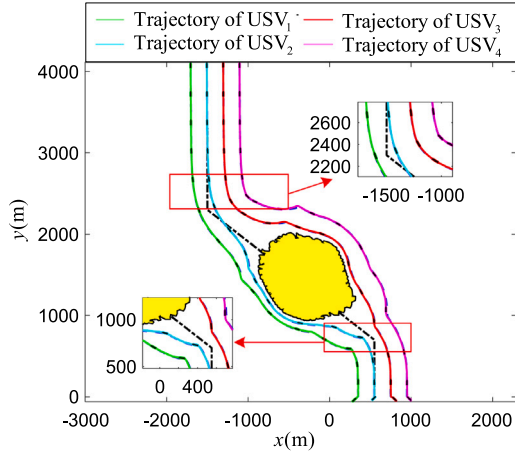
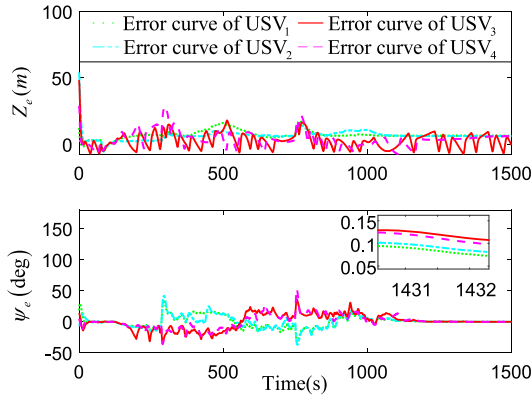


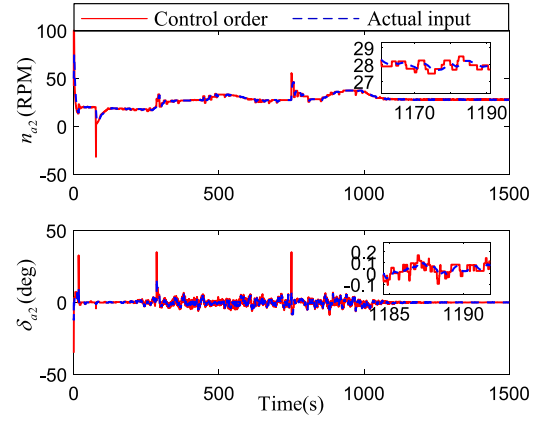
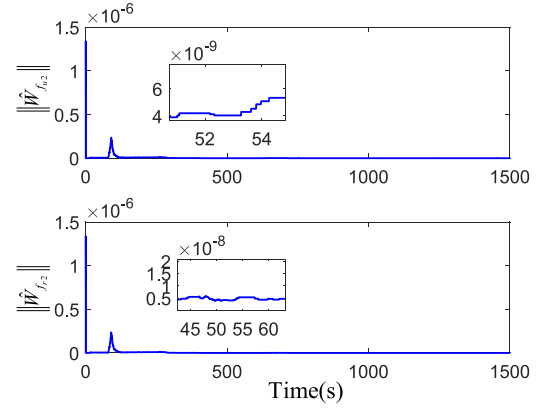
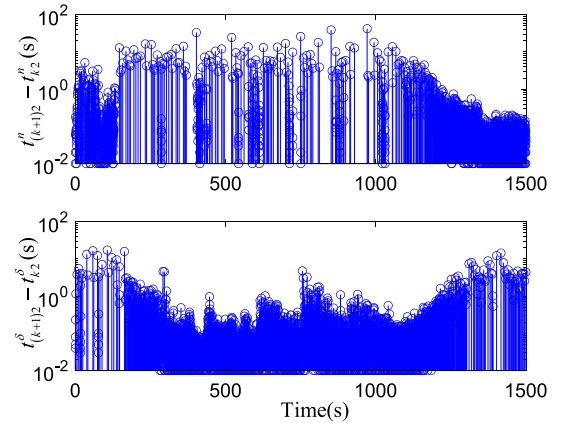
Fig. 3. Obstacle avoidance trajectories of the USVs.

Fig. 4. Position error Z_e and heading angle error ψ_e between each follower and the virtual leader.

corresponding reference trajectory is exhibited in Fig. 4. It is obvious that the tracking error converges to a smaller value. Fig. 5 shows that the stair-stepping control orders n_{a2} and δ_{a2} are generated by controllers, which indicate that the actuators only execute the control orders at the triggering time. The rudder servo system converts the reference signals to actual inputs n_{a2} and δ_{a2} , which is consistent with engineering practice. It is obvious that, unlike most ETC strategies, the proposed DETC algorithm can enable the NNs weight estimators to be updated at triggering instants. Fig. 6 indicates the norm of NNs' weight estimators under the developed event-triggered rules, presenting that this approach can reduce computational burden. From Fig. 7, the interval conditions between two adjacent event-triggered instants are shown. It is further demonstrated that the computational load and wear from the controllers to the actuators is alleviated, improving the control performance of the system.

5.2. Comparison example

In Section 5.1, the DETC is proposed to simulate the navigation of USVs. In order to verify the superiority of this control algorithm in terms of computational burden and actuator wear, we select one of the USVs in Section 5.1 to indicate the main results more clearly, without loss of generality. The main comparison results are exhibited in Figs. 8–10. Fig. 8 presents a comparative analysis of trajectory tracking among the continuous algorithm, the DETC algorithm, the proposed algorithm without fault-tolerant and the one in Ma et al. (2020). Although each algorithm achieves acceptable tracking accuracy, the proposed algorithm

Fig. 5. The control order and actuator input of USV₂.Fig. 6. Norm of the NNs weight estimator $\|\hat{W}_{f_{a2}}\|$, $\|\hat{W}_{f_{r2}}\|$.Fig. 7. Interval conditions between two adjacent event-triggered instants for propeller speed and rudder angle for USV₂.

in this paper shows a marked improvement in performance over Ma et al. (2020). It is noted that the continuous scheme exhibits better control performance than the DETC algorithm. Additionally, it can be seen that the actuator fault is introduced between 250 to 300 s. Although the actual input of the USV is only 80% effective due to the impact of the PLOE, the actual control command is still able to maintain the state before the fault occurred, which is attributed to the designed fault-tolerant control strategy. From the comparison of position error and course angle error in Fig. 9, it can be seen that due to the introduction of DETC, the swing amplitude of the rudder angle

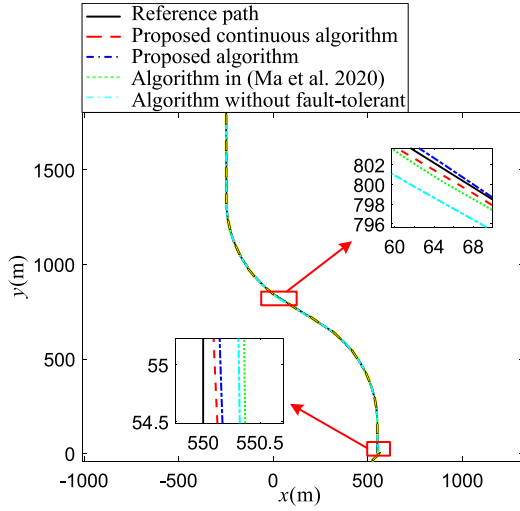


Fig. 8. Comparative tracking performance under different control algorithms.

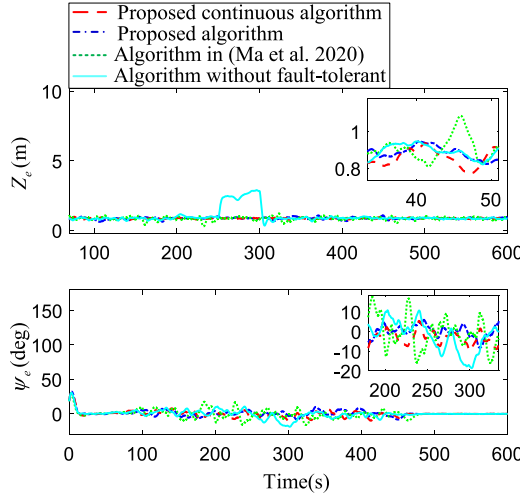


Fig. 9. Tracking position error z_e and heading angle errors ψ_e : proposed algorithm in this paper, the compared algorithm (Ma et al., 2020).

controlled by DETC is reduced and the calculation load is decreased. Fig. 10 describes the comparative effect of the control input given by the four methods. It is apparent that the control orders of the proposed algorithm change slowly and chatter less than the compared results.

For the quantitative purpose, three popular performance specifications (Zhang et al., 2024b) are used to evaluate the aforementioned comparison. They are the mean absolute error (MAE), the mean absolute control input (MAI) and the mean total input variation (MTV), which describe the responsiveness of system, the energy consumption and the smoothness properties of the controllers. The quantitative comparison effect can be intuitively illustrated in Fig. 11. As can be observed from the comparison result, although the MAE of value ψ_e in the proposed algorithm is larger than that in Ma et al. (2020), the other values are lower than those in Ma et al. (2020). Therefore, the quantitative comparison demonstrates that the proposed algorithm has significant advantages in terms of higher tracking accuracy and lower energy consumption.

6. Conclusion

An improved APF method for cooperative USV formation obstacle avoidance has been proposed in this paper. Considering the practical

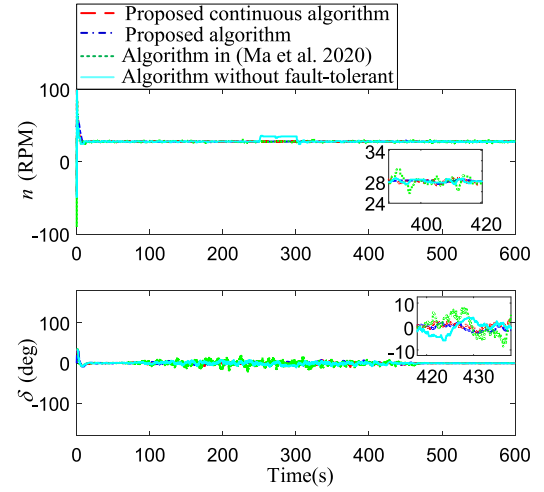


Fig. 10. control input n_a and δ_a : Proposed algorithm in this paper, the compared algorithm (Ma et al., 2020).

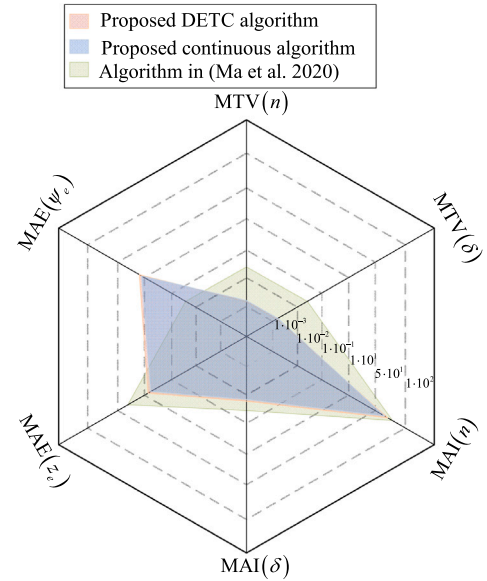


Fig. 11. Quantitative comparison radar chart.

requirements, the proposed guidance principle could guide the formation to carry out the automatic obstacle avoidance mission along the predefined path. In addition, a dynamic event-triggered cooperative formation control algorithm considering actuator faults is proposed by employing dynamic event-triggered control, backstepping approach and RBF-NNs technique. The proposed guidance algorithm, although it enables fast response and formation reconfiguration of USV formations, does not adequately consider whether the planned path is the optimal path, i.e., the influence of interval. In addition, the numerical example does not consider the impact of moving vessels on engineering practice, which will be considered by the authors in the next work. Last but not least, the authors should further discuss and propose high precision control algorithms for external disturbances and model uncertainties. Flexible obstacle avoidance for moving obstacles and static obstacles is one of the future research goals. Future work needs to continuously optimize formation obstacle avoidance schemes.

CRediT authorship contribution statement

Guoqing Zhang: Writing – original draft, Software, Project administration, Methodology, Formal analysis. **Chengqian Shi:** Writing – review & editing, Validation, Software, Conceptualization. **Jiqiang Li:** Writing – review & editing, Validation, Formal analysis. **Xianku Zhang:** Writing – review & editing, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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